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EDUCATION

- TU Munich, TU Wien, TU Dresden, and the University of Twente** Oct 2025 - Present
Erasmus Mundus Joint Masters in Cartography - European Union Scholar
- University of Mines and Technology (UMaT)** September 2019 - September 2023
BSc Geomatic Engineering **First Class Honours (81.88, GPA 3.8/4)**
BSc Thesis: "Data Quality Assessment of Geodetic Observations Using Statistical Testing Procedures"
Advisor: Prof. Yao Yevenyo Ziggah.

ONLINE COURSES

- Machine Learning and Deep Learning Bootcamp in Python **June 2026 - Present**
- AI Fluency: Framework & Foundations **June 2026**
- Neural Networks and Deep Learning **September 2025**
- Data Analysis with R programming **April 2024**
- Process Data from Dirty to Clean **March 2024**
- Prepare Data for Exploration **January 2024**

RESEARCH EXPERIENCE

- Evaluating UN-OCHA Natural Disaster Map Icons Under Time Pressure and Contextual Framing** May 2026 - Present
Manuscript in Preparation for Peer-Reviewed Submission (Co-author)
The study investigates how effectively the UN-OCHA icon set is recognized under time pressure and within a real-world scenario.
- Spatial Hotspots of Mpox Risks in Democratic Republic of the Congo** Oct 2025 - Feb 2026
Geo-Health Project (Supervisor: Prof Justine Blanford)
Combined weighted risk criteria from multiple geospatial datasets using a multi-criteria decision analysis (MCDA) approach to map mpox-risk zones across 19 health zones and ran spatial data quality checks to support public-health resource allocation.
- SAR-Based Land-Cover Classification Using CV-CNN** Oct 2025 - Feb 2026
Big Geospatial Data Project (Supervisor: Dr Kwasi Nyako Poku-Agyemang)
Contributed to automated land-cover classification of agricultural land into 15 classes using SAR data. Developed data processing pipelines and compared complex-valued CNN (CV - CNN) with conventional CNN approaches. CV - CNN attained 93.21% overall accuracy compared to 92.14% for RV-CNN (Cohen's $\kappa = 0.942$ vs. 0.928) with largest gain observed for crops such as rapeseed (+18.83% in F1-score).
- Data Quality Assessment of Geodetic Observations Using Statistical Testing Procedures. (January 2023 - August 2023)**
Geomatic Engineering undergraduate thesis (Supervisor: Prof Yao Yevenyo Ziggah)
The research focused on identifying outliers in geodetic observations using robust estimation methods such as Danish, RANSAC algorithm and Data Snooping Procedures. The results showed that all the robust estimation methods can detect and handle outliers in observations with satisfactory accuracy level. However, Random Sample Consensus (RANSAC) algorithm conceded the greatest number of outliers.

- **Analysis of Land Use Land Cover Change for Tarkwa-Nsuaem District - Ghana (March 2023 - April 2023)**

Urban Planning Project for Tarkwa-Nsuaem District (Supervisor: Dr. Micheal Aduah)

The research captured the land Use and Land Cover (LULC) changes in some areas in Tarkwa-Nsuaem District. The research predicted a ten-year LULC analysis for the selected areas.

HONOURS AND AWARDS

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| • Erasmus Mundus Scholarship Award , European Commission | 2025 - Present |
| • Best Poster Presentation Award , Technical University of Munich | 2025 |
| • Best Female Graduating Student , Geomatic Eng Department, UMaT | 2023 |
| • Beneficiary , Total Ghana Scholarship | 2021 - 2023 |
| • Departmental Excellence Award , UMaT | 2022 |
| • Beneficiary , Ghana Chamber of Mines Tertiary Education Fund | 2019 - 2021 |

TEACHING AND WORK EXPERIENCE

Teaching and Research Assistant UMaT Geomatic Department-Ghana **Nov 2023 - Sept 2025**

- Supervised undergraduate students to successfully complete their field and research work
- Grouped students to work on projects and present findings
- Organized tutorial sessions for undergraduate students in GIS, Remote Sensing and Cartography
- Assisted my supervisor with research (GNSS receivers' operation, data collection and surveying)
- Marked and graded students' assignments and exams script

Intern, Achiever's Geo-Consult, Tarkwa **October 2022 - January 2023**

- Performed Volume computation of a mining company using Geovia Surpac
- Assisted in Setting out and demarcation of parcel of land
- Assisted in Real Estate Development and Resettlement of communities

Intern, Geo-Tech Surveys, Ringway Estate - Osu **June 2022 - August 2022**

- Conducted cadastral surveys using GNSS/RTK for land registration and parcel demarcation
- Undertook Geo-Technical projects in underground structures and transportation networks
- Performed General and Engineering Surveying

Intern, Geo-Tech Systems, Samora Machel Rd- Accra **October 2021 - December 2021**

- Calibrated survey instruments such as theodolite from Topcon, Sokkia and Trimble
- Assisted in the distribution of sophisticated and cutting-edge positioning instruments and accessories
- Assisted in the repairing and maintenance of positioning instruments.

SEMINARS AND CONFERENCES

- **International Federation of Surveyors (FIG)** - Your World, Our World: Resilient Environment and Sustainable Management for All. **May 2024**
- **FIG Climate Compass Task Force Seminar** - How to Use a Range of Survey Tools to Monitor Flooding from Tropical Cyclones **March 2024**
- **Ghana Geospatial Society Annual conference** - Harnessing Geospatial Technology for the Achievement of the SDGs in Ghana **November 2023**
- **Geovia User Conference** - Dassault Systemes upgrade on Surpac Software **July 2023**
- **GESA Seminar** - The Use of Drones in Geomatics **November 2022**
- **AusiMM Tarkwa Student Chapter Conference** - Bridging the Gap Between Academia and Industry - A Catalyst for a Skilled Workforce **September 2022**

PRESENTATIONS

- **Poster Presentation, Technical University of Munich** **2025**
Presented on: "SAR-Based Land-Cover Classification Using CV-CNN" – TUM, Germany
- **Geomatic Engineering Department, Annual Synopsis Seminar** **2023**
Presented on: "Data Quality Assessment of Geodetic Observations Using Statistical Testing Procedures"

MEMBERSHIPS

- International Society for Photogrammetry and Remote Sensing **2024 - Present**
- Ghana Geospatial Society, Ghana **2023 - Present**
- Australasian Institute of Mining and Metallurgy (AusIMM), Ghana **2020 - 2023**
- Women in Engineering (WINE), UMaT, Ghana **2019 - 2023**

SOFTWARE

- Microsoft Office Suite (Word, Excel, PowerPoint), MATLAB, JavaScript
- Geovia Surpac, ArcGIS Pro, AutoCAD, QGIS, Python, SatScan, GeoDa and R
- Leaflet.js, MapLibre GL JS, Maputnik, and Affinity Designer

OTHER SKILLS AND COMPETENCIES

- Excellent team player
- Critical analyst and very innovative
- Easily adaptable to changes and readiness to tackle new problems

LANGUAGE

- Good English communication skills (Distinguished levels in Listening, Speaking, Reading and Writing)
- French and German at Elementary level

LEADERSHIP EXPERIENCE

- **Assistant Ambassador - Innovation Fair** **2023**
- **Class Representative** **2022 - 2023**
- **Ladies Wing Vice President:** UMaT **2022 - 2023**
- **Mentor** - Geomatic Engineering Student Association, UMaT **2022**
- **Main Women Commission Organizer:** UMaT **2021 - 2022**
- **Course Representative** - Geomatic Engineering Department **2021**

COMMUNITY/VOLUNTEERING SERVICES

Student Teaching Assistant, UMaT, Sekondi-Tarkwa **November 2023 - September 2025**

- Conducted free tutorial sessions for first and second-year undergraduate students
- Assisted lecturers in research and laboratory works
- Marked and graded students' scripts

International Federation of Surveyors (FIG) Working Week **May 2024**

- Assisted with participant registration, welcoming attendees, and providing directions at the conference.
- Supported the setup and preparation of presentation venues and conference facilities.
- Assisted in the distribution of conference materials and souvenirs to participants.

Students in Community Services **September 2022**

- Provided clothing and healthcare to underserved communities in Ghana's Western and Central Regions.
- Conducted teaching and training exercises for the students in the communities

REFERENCES

Available upon request